Connor Johnston

Gainesville, FL | cjohnston1@ufl.edu | +1.352.988.9397

Technical Skills

- **Programming:** Python, C++
- **Optimization software:** Pyomo, Gurobi, HiGHS
- **Methods:** Monte Carlo methods, mixed-integer programming, linear programming, disjunctive cutting-plane methods, algorithm design, computational experiments and benchmarking
- **Development tools:** Git/GitHub

Education

University of Florida

- Ph.D. Industrial and Systems Engineering — Expected December 2027
- M.S. Industrial and Systems Engineering — Expected December 2027

Florida State University

- B.S. Applied and Computational Mathematics with Honors — 2019–2023
- B.A. Philosophy — 2019–2023

Research and Software Development Experience

Sandia National Laboratories — May–August 2026

Summer Research Intern, Computer Science Research Institute (CSRI)

- Implemented disjunctive cutting-plane functionality in Python for Pyomo, an open-source optimization modeling package.
- Tested the implementation and prepared the contribution for integration into Pyomo; merge pending.

Preprints

[1] **C. R. Johnston** and A. M. Kazachkov. “Symmetrically Fair Allocations of Indivisible Goods.” 2024. Revising for submission. Preprint

Working Papers

[2] C. Chen, **C. R. Johnston**, D. R. Moran, and A. M. Kazachkov. “Scaled Simple Disjunctive Cuts.” Completed manuscript.

[3] **C. R. Johnston**, S. Davis, E. Johnson, and A. M. Kazachkov. “Simple Disjunctive Cuts for Cardinality Constrained Programs.” Completed manuscript.

Undergraduate Thesis

[4] **C. R. Johnston**. “A Novel Monte Carlo Algorithm for Pricing Barrier Options.” 2022. Thesis

Awards

- Honorable Mention Award, Mixed Integer Programming Workshop, 2026.
- Student Organization Leadership Award, University of Florida Industrial and Systems Engineering, 2025 and 2026.
- Dean’s Research Award, University of Florida Industrial and Systems Engineering, 2023–Present.
- Kenneth G. Boback Award, Florida State University Department of Mathematics, 2023.

Leadership and Service

- INFORMS Student Chapter Talk Coordinator — 2025–2026
- INFORMS Student Chapter President — 2024–2025

Presentations

Invited Talks

- Discrete Optimization Talks (DOTs), “The many facets of simple disjunctions,” September 11, 2026.
- INFORMS Annual Meeting, “Symmetrically Fair Allocations of Indivisible Goods,” October 2024.
- American Mathematical Society, “A Novel Monte Carlo Algorithm for Pricing Barrier Options,” March 2023.

Posters

- Mixed Integer Programming Workshop, “An output sensitive polynomial time algorithm for enumeration of the vertices of the reverse polar set related to simple disjunctions,” 2026.
- Mixed Integer Programming Workshop, “Simple Disjunctive Cuts are All We Need!” June 2025.
- Mixed Integer Programming Workshop, “Symmetrically Fair Allocations of Indivisible Goods,” June 2024.